The Master Quadratic of Discrete Spacetime: Deriving α^{-1} from Lattice Gauge Theory

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We derive the master quadratic $x^2 - 16(G^*)^2 x + 16(G^*)^3 = 0$ from the lattice action via Dyson self-consistency, where $G^* = \sqrt{2}\,\Gamma(1/4)^2/(2\pi) \approx 2.9587$ is the lemniscatic constant. The derivation proceeds through six steps: lattice action, Gauss constraint, 16 degrees of freedom on the minimal 2^3 cell, one-loop effective action, vacuum polarization, and self-consistency. The roots $x_+ = 137.036$ and $x_- = 3.024$ match α^{-1} (1.26 ppm) and $N_c = 3$ (0.8%). We present a computational ontology distinguishing detection (1 logic gate), measurement (≥ 2 gates, inorganic), and observation (≥ 2 gates, organic). Wave function "collapse" is Bayesian updating by systems with ≥ 2 gates; systems with 0 gates (rocks, corpses) cannot construct wave functions to collapse. The quadratic is a structural property of discrete geometry requiring no observer.

INTRODUCTION

The fine structure constant $\alpha \approx 1/137$ lacks theoretical derivation. We demonstrate it emerges as a self-consistency condition of lattice gauge dynamics on discrete spacetime.

A persistent confusion concerns the observer's role in quantum mechanics. We propose a **computational ontology**:

Category	Gates	Substrate	Capability
Dead	0	None	None
Brain Dead	1	Any	Detection
Measurer	≥ 2	Inorganic	Inference
Observer	≥ 2	Organic	Inference

Detection (1 gate): Boolean yes/no. A Geiger counter clicks but does not "know" it detected anything.

Measurement (≥ 2 gates): Comparison across time. Memory. Inference. Model-building.

The physics is identical regardless of gate count. The lattice dynamics proceed whether 0 or 10^{100} gates exist.

TIME AS EMERGENT

Consider a video showing a stationary white dot. Is there informational difference between frame 1 and frame 10^{18}? No. If states are identical at t_1 and t_2, no computation distinguishes them. **Time is emergent**, secondary to existence itself.

A "tick" is dimensionless discrete evolution. Sans time, a voxel is 1D information: exists or not. Physics emerges when $t > 1$.

THE WAVE FUNCTION

A particle is not "in all places at once." At any tick t_i, it has definite location. The wave function is the **aggregate over time**, constructed by systems with ≥ 2 gates.

A rock (0 gates) cannot collapse wave functions because it cannot construct them. "Collapse" is Bayesian updating, not physics.

THE LATTICE ACTION

On a cubic lattice $\mathbf{L} \subset \mathbb{Z}^3$:

$$S[\mathbf{J}] = \sum_t \sum_v \left[\frac{1}{2}|\partial_t \mathbf{J}|^2 - \frac{1}{2}|\nabla \mathbf{J}|^2\right] \tag{1}$$

where $\mathbf{J}(v,t) \in \mathbb{R}^3$ is the flux field. No observer terms appear.

GAUSS CONSTRAINT

Theorem 1: Any $\mathbf{J}$ decomposes as $\mathbf{J} = \mathbf{J}_T + \mathbf{J}_L$ with $\nabla \cdot \mathbf{J}_T = 0$ (transverse) and $\mathbf{J}_L = \nabla\phi$ (longitudinal).

The constraint $\nabla \cdot \mathbf{J} = 0$ eliminates longitudinal modes. On a compact periodic lattice, $\nabla^2\phi = 0$ implies $\phi =$ const, so $\mathbf{J}_L = 0$.

DEGREE OF FREEDOM COUNTING

Theorem 2: On the $2 \times 2 \times 2$ lattice:

$$N_{\mathrm{DoF}} = 24 - 7 - 1 = 16 \tag{2}$$

counting flux components (8×3) minus Gauss constraints $(8-1)$ minus gauge freedom.

The number 16 admits four derivations: 4^2, 2^4, $24-8$, and $32/2$.

ONE-LOOP EFFECTIVE ACTION

Theorem 3: Integrating out transverse fluctuations:

$$\Gamma_{\text{1-loop}}(x) = \frac{1}{2}\text{Tr}\ln K(x) \tag{3}$$

with $K(x) = \omega^2 + x|\mathbf{k}|^2$. The trace runs over 16 modes.

VACUUM POLARIZATION

Theorem 4: The polarization correction is:

$$\Pi(x) = \frac{16(G^*)^3}{x} \tag{4}$$

where factors arise from: 16 (mode count), $(G^*)^3$ (lemniscatic regularization), $1/x$ (propagator).

The lemniscatic constant:

$$G^* = \frac{\sqrt{2}\,\Gamma(1/4)^2}{2\pi} = 2.9586751... \tag{5}$$

is uniquely selected by complex multiplication at $j = 1728$.

THE MASTER QUADRATIC

Self-consistency requires $x = 16(G^*)^2 - \Pi(x)$:

$$x = 16(G^*)^2 - \frac{16(G^*)^3}{x} \tag{6}$$

Rearranging:

$$\boxed{x^2 - 16(G^*)^2 x + 16(G^*)^3 = 0} \tag{7}$$

The roots:

$$x_+ = 137.0358... \tag{8}$$
$$x_- = 3.0236... \tag{9}$$

Conjecture: $x_+ = \alpha^{-1}$ and $\lfloor x_- \rfloor = N_c = 3$.

TABLE I. Comparison with experiment

Quantity	Predicted	Experimental	Error
α^{-1}	137.0358	137.0360	1.26 ppm
N_c	3.024	3	0.8%

BARYON MASSES

The composition constant $K_{\text{comp}} = m_e/\pi$ represents 3D manifestation cost:

$$M_p = \left(\frac{13}{\alpha} + 55\right) m_e - \frac{m_e}{\pi} = 938.272 \text{ MeV} \tag{10}$$

$$M_n = M_p^{\text{geo}} + (\phi^2 - 12\alpha)m_e - \frac{m_e}{\pi} = 939.565 \text{ MeV} \tag{11}$$

$$M_\Delta = \left(\frac{17}{\alpha} + 81\right) m_e - \frac{m_e}{\pi} = 1231.7 \text{ MeV} \tag{12}$$

TABLE II. Baryon mass predictions

Particle	Predicted	Experimental	Error
M_p (MeV)	938.2724	938.2720	0.4 keV
M_n (MeV)	939.5654	939.5654	< 1 eV
M_Δ (MeV)	1231.7	1232	0.03%

THE MEASUREMENT PROBLEM DISSOLVED

There is no measurement problem:

1. At any tick, particles have definite states
2. "Superposition" describes model uncertainty (≥ 2 gates)
3. "Collapse" is updating models upon new data
4. Physics proceeds regardless of gate count

Death is the transition from ≥ 2 gates to 0 gates. A corpse is computationally equivalent to a rock.

CONCLUSION

The master quadratic $x^2 - 16(G^*)^2 x + 16(G^*)^3 = 0$ is **derived**, not postulated. It produces α^{-1} to 1.26 ppm and N_c to 0.8%.

The derivation is observer-free. Fundamental constants are structural properties of discrete geometry. Wave functions are epistemic constructs requiring ≥ 2 logic gates. A rock cannot collapse wave functions because a rock cannot construct them.

The quadratic requires no observer to be true.

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